Scanning Electron Microscope (SEM)

Eka Nurfani

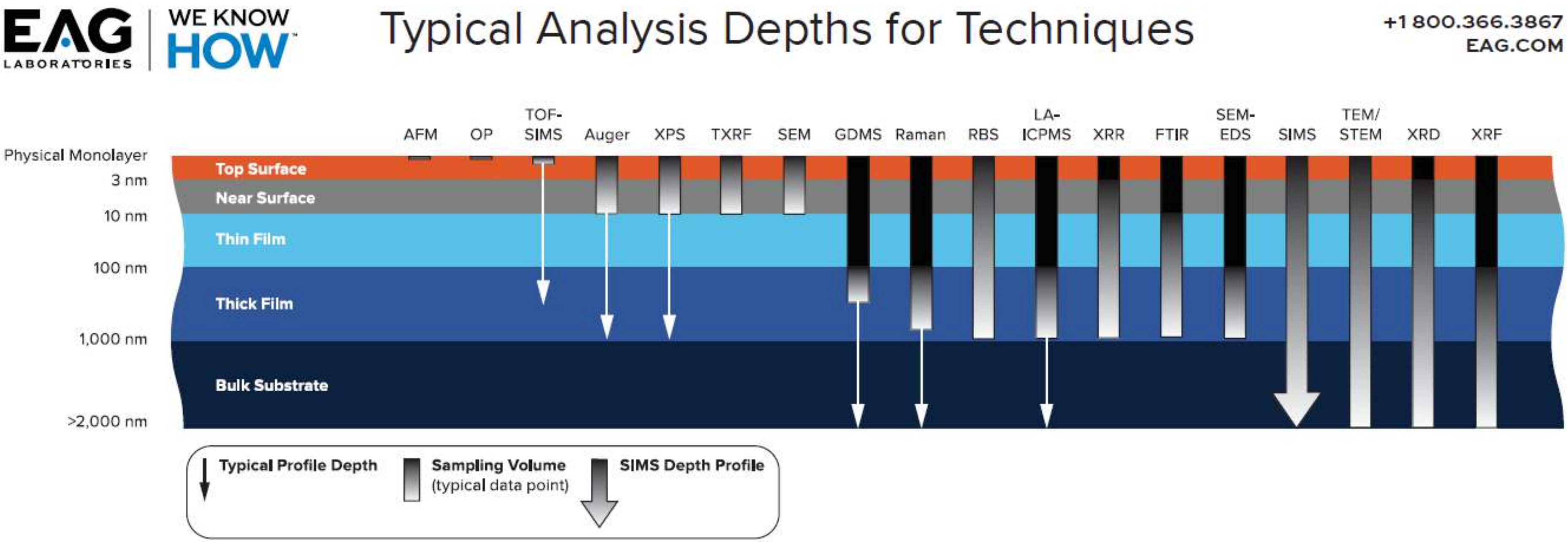
Disampaikan pada **MK Karakterisasi Material Lanjut** (MT3015)

Minggu ke-9

Outline

- What can we use a SEM for?
- How do we get an image?
- Electron beam-sample interactions
- Signals that can be used to characterize the microstructure
  - Secondary electrons
  - Backscattered electrons
  - X-rays
- Components of the SEM
- Some comments on resolution

Characterizations vs Depth



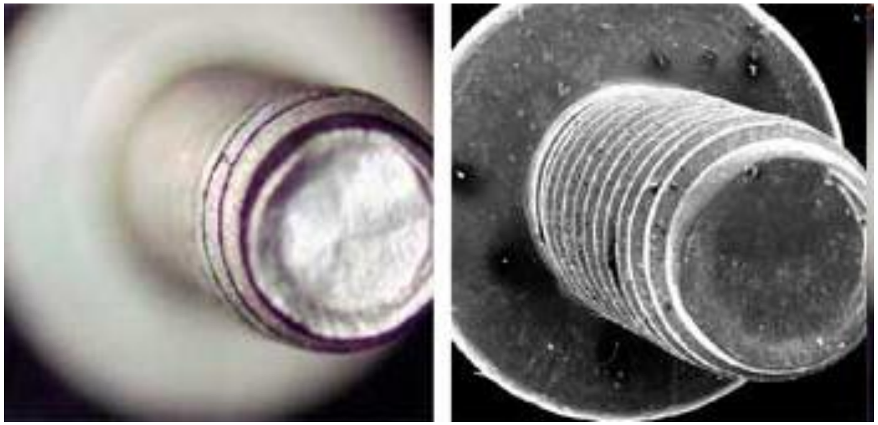
https://engineering.virginia.edu/

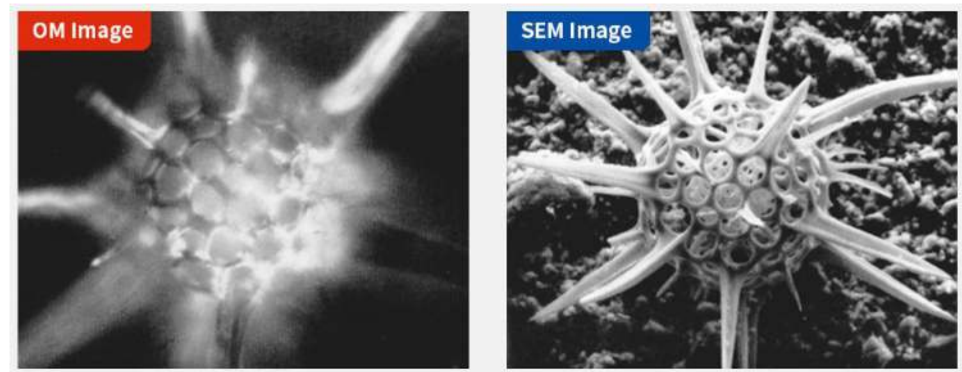
What can we study in a SEM?

- Topography and morphology
  - **Morphology** studies the shape, texture and distribution of materials at a surface.
  - **Topography** focuses on the quantitative dimensional measurement of features on a surface.
- Compositional
- In-situ experiments:
  - Reactions with atmosphere
  - Effects of temperature

Depth of focus

Optical microscopy vs SEM



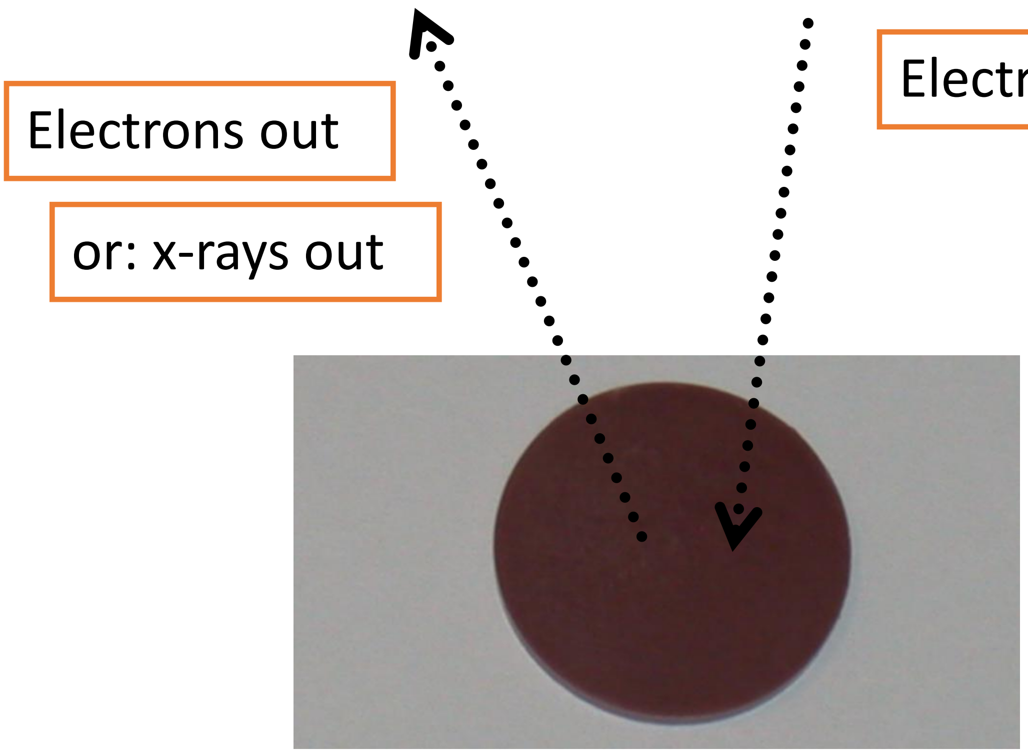


Screw length: ~ 0.6 cm

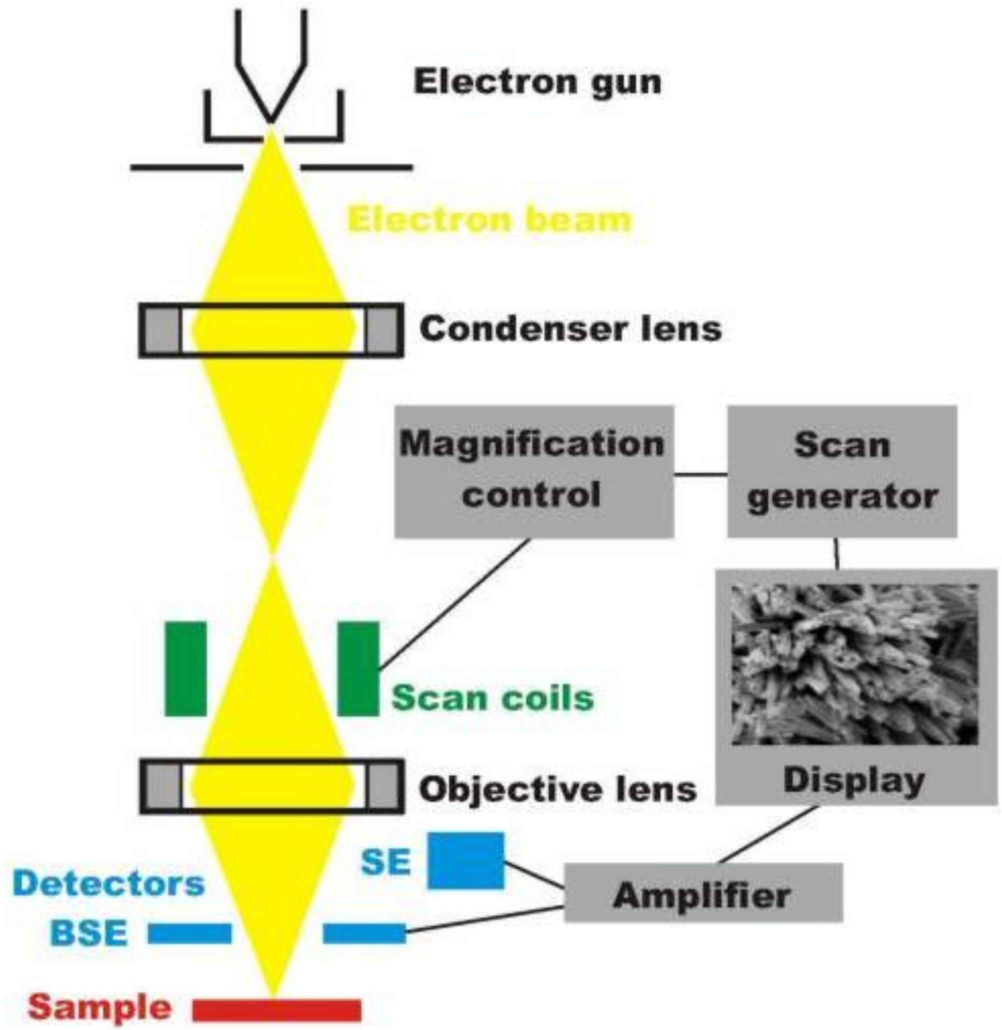
- A SEM typically has better depth of focus than a optical microscope, making SEM suitable for studying rough surfaces
- The higher magnification, the lower depth of focus

How do we get an image?

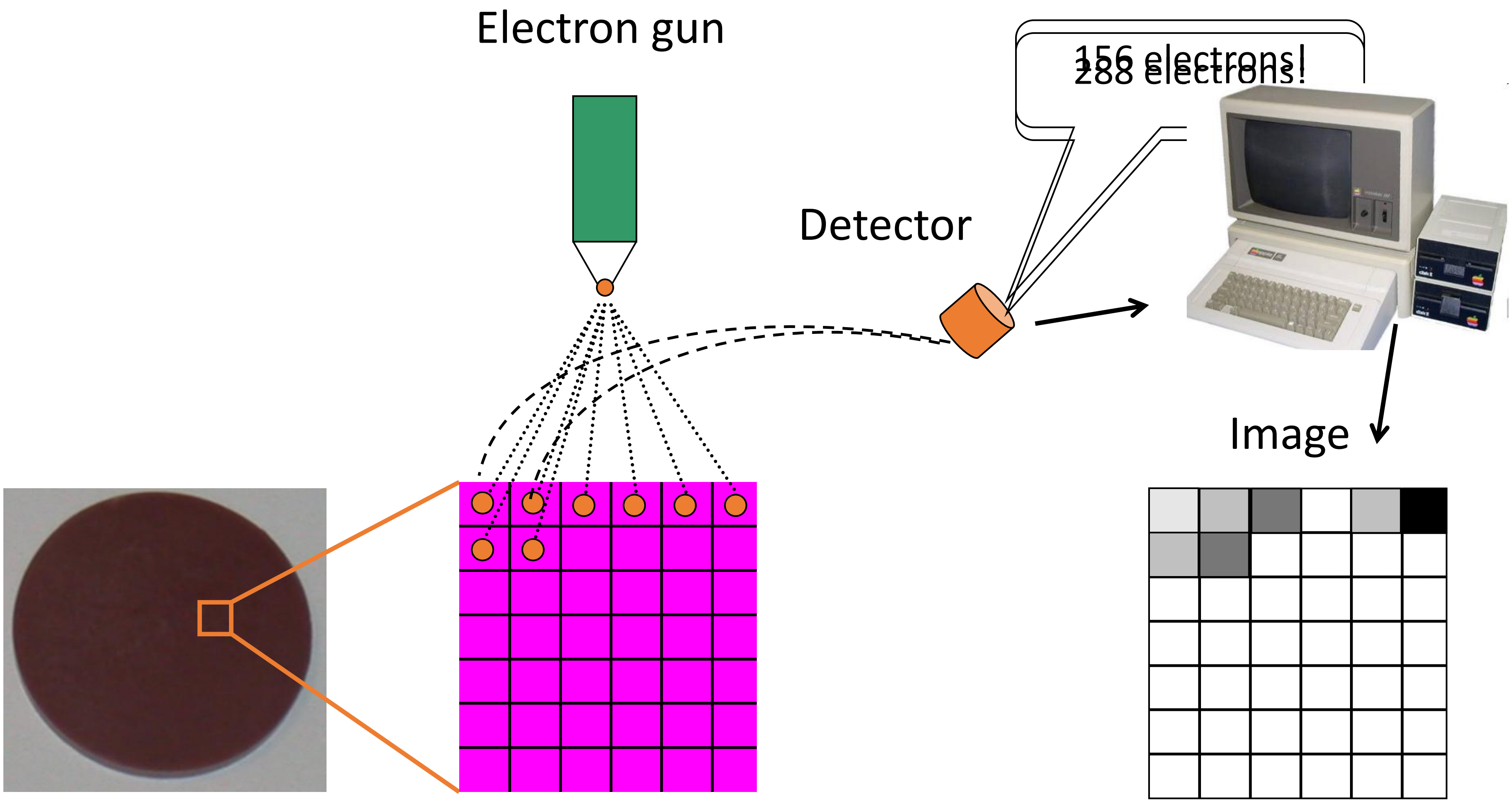
- We shoot high-energy electrons and analyze the outcoming electrons/x-rays



The instrument in brief

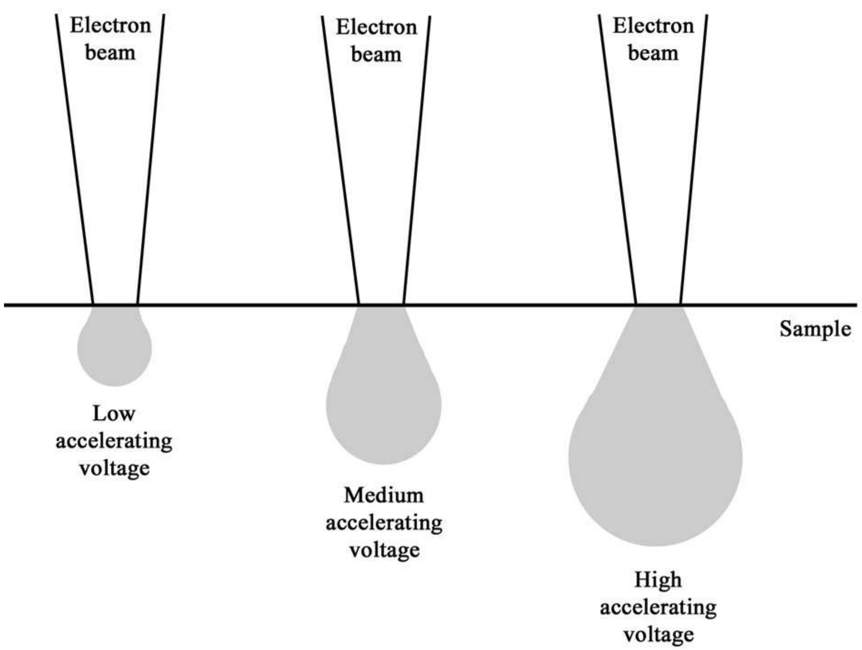


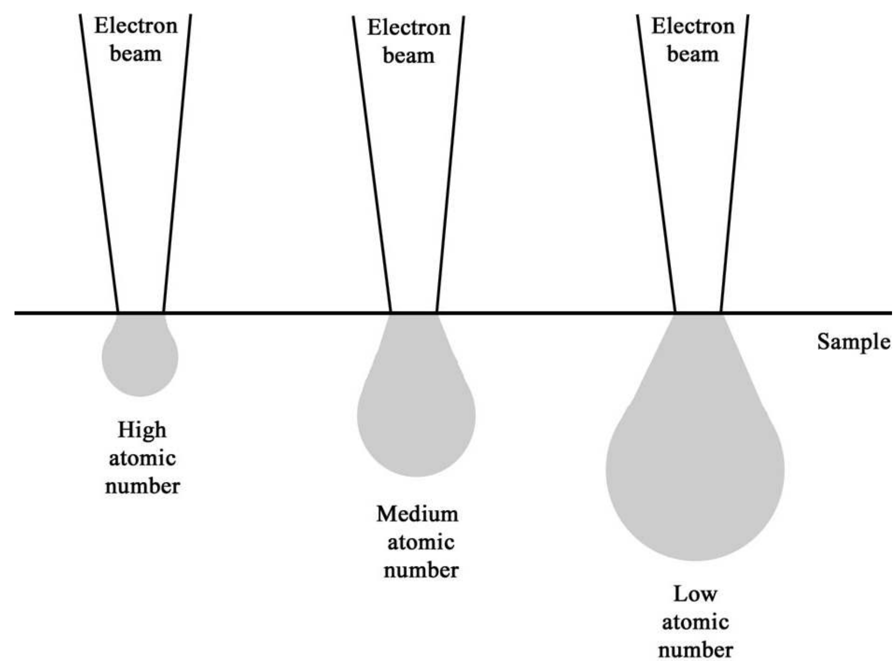
How do we get an image?



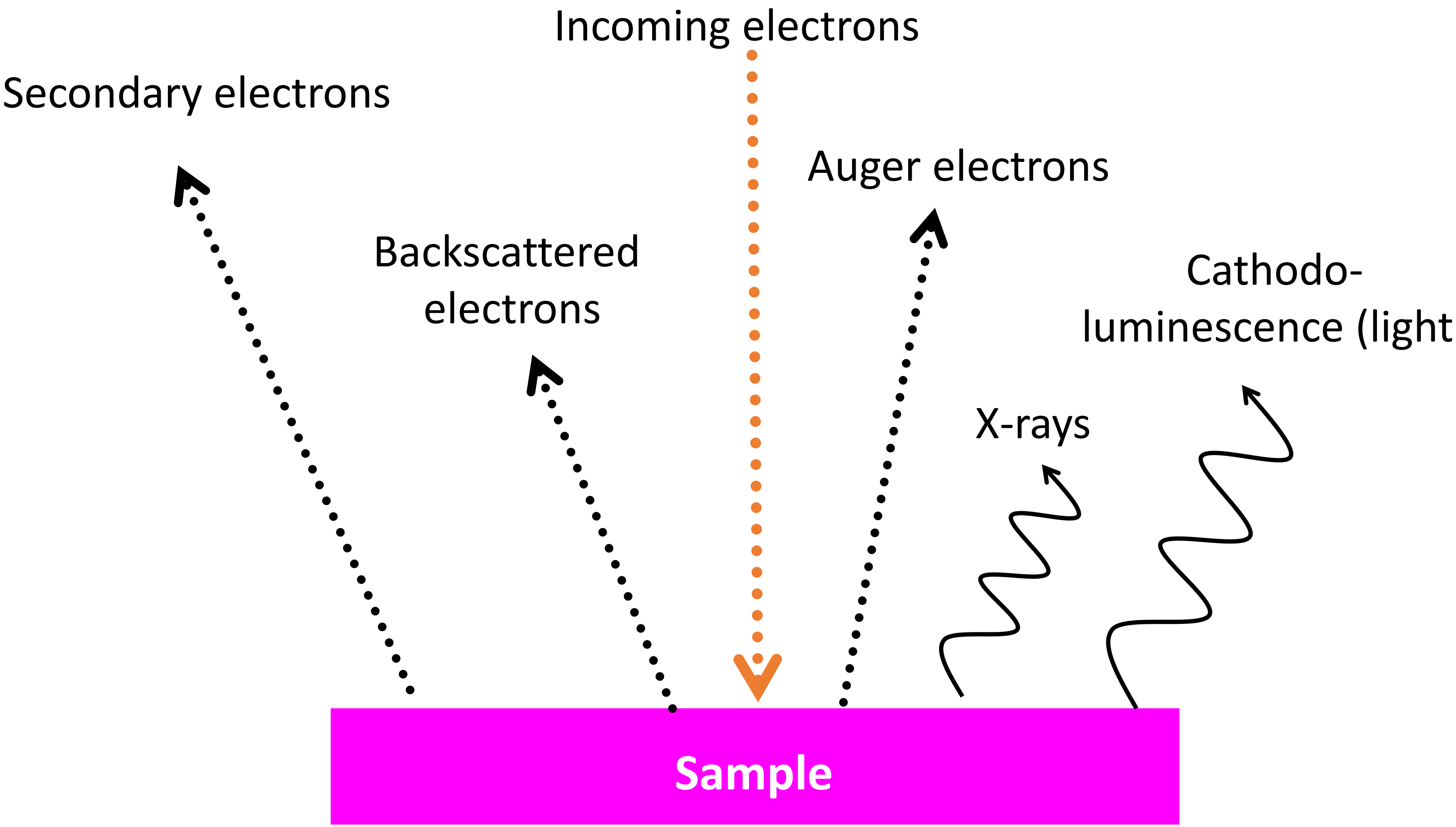
Electron beam-sample interactions

- The incident electron beam is scattered in the sample, both elastically and inelastically
- This gives rise to various signals that we can detect
- Interaction volume increases with increasing acceleration voltage and decreases with increasing atomic number





Signals from the sample

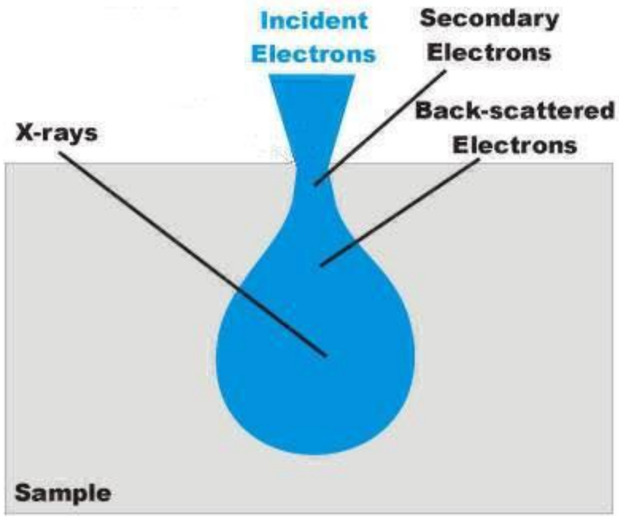


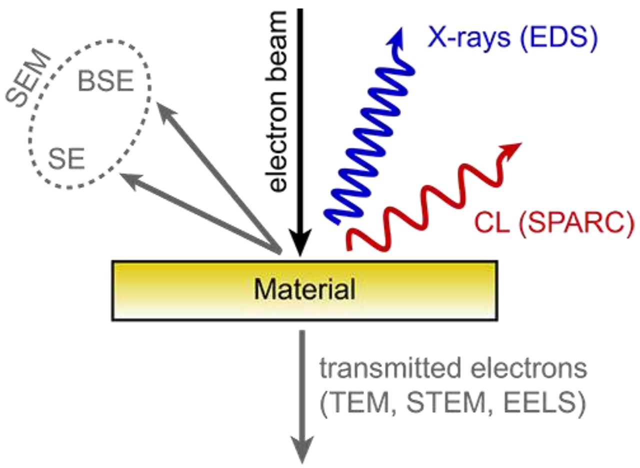
Electron beam-sample interactions

- **Backscattered electrons** – these are high energy electrons which are scattered out of the specimen, losing only a small amount of energy.

- **Secondary electrons** – these originate in the specimen itself, and have a much lower energy than the BSE (typically < 50 eV).

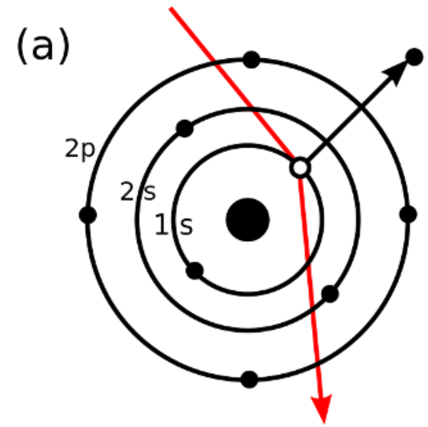
- **X-rays** - These give information about the elemental composition of the sample.





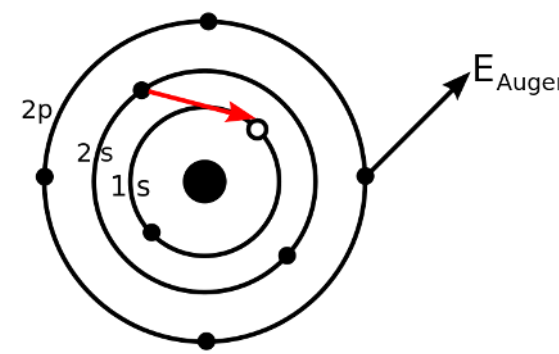
Electron beam-sample interactions

- SEM-**CathodoLuminescence** → emission of photons of characteristic wavelengths from a material that is under high-energy electron bombardment produced in a scanning electron microscope.



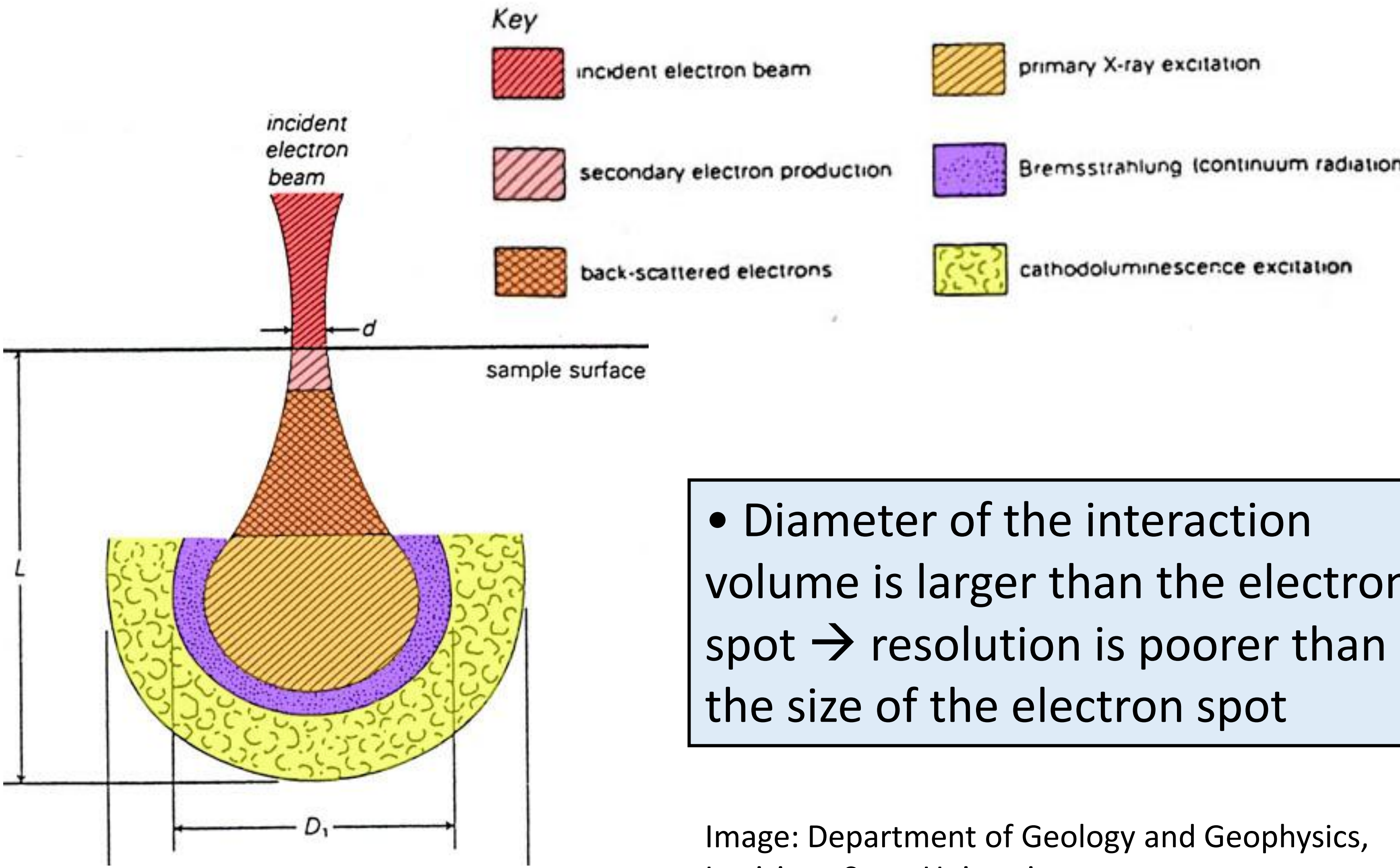
Electron collision

- **Auger electrons** → electrons that are emitted when an electron from a higher energy level falls into a vacancy in an inner shell.



Auger electron emission

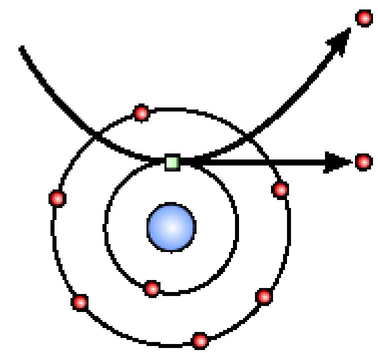
Where does the signals come from?



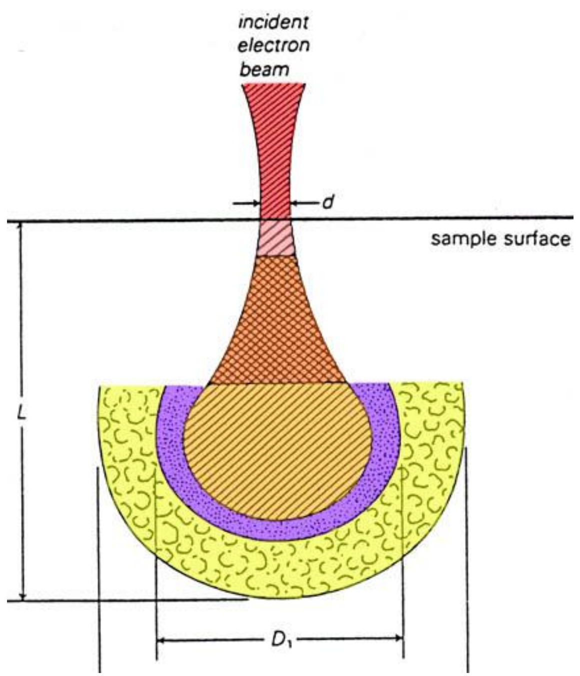
- Diameter of the interaction volume is larger than the electron spot → resolution is poorer than the size of the electron spot

Image: Department of Geology and Geophysics, Louisiana State University

Secondary electrons (SE)

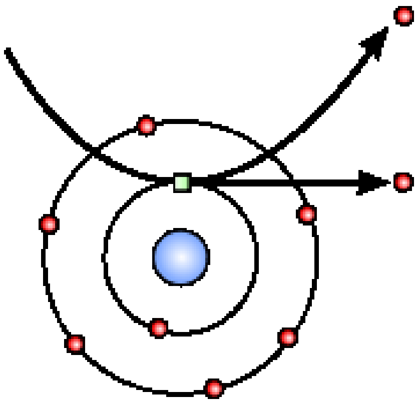


- Generated from the collision between the incoming electrons and the loosely bonded outer electrons
- Low energy electrons (~10-50 eV)
- Only SE generated close to surface escape (topographic information)
- Number of SE is greater than the number of incoming electrons
- We differentiate between SE1 and SE2



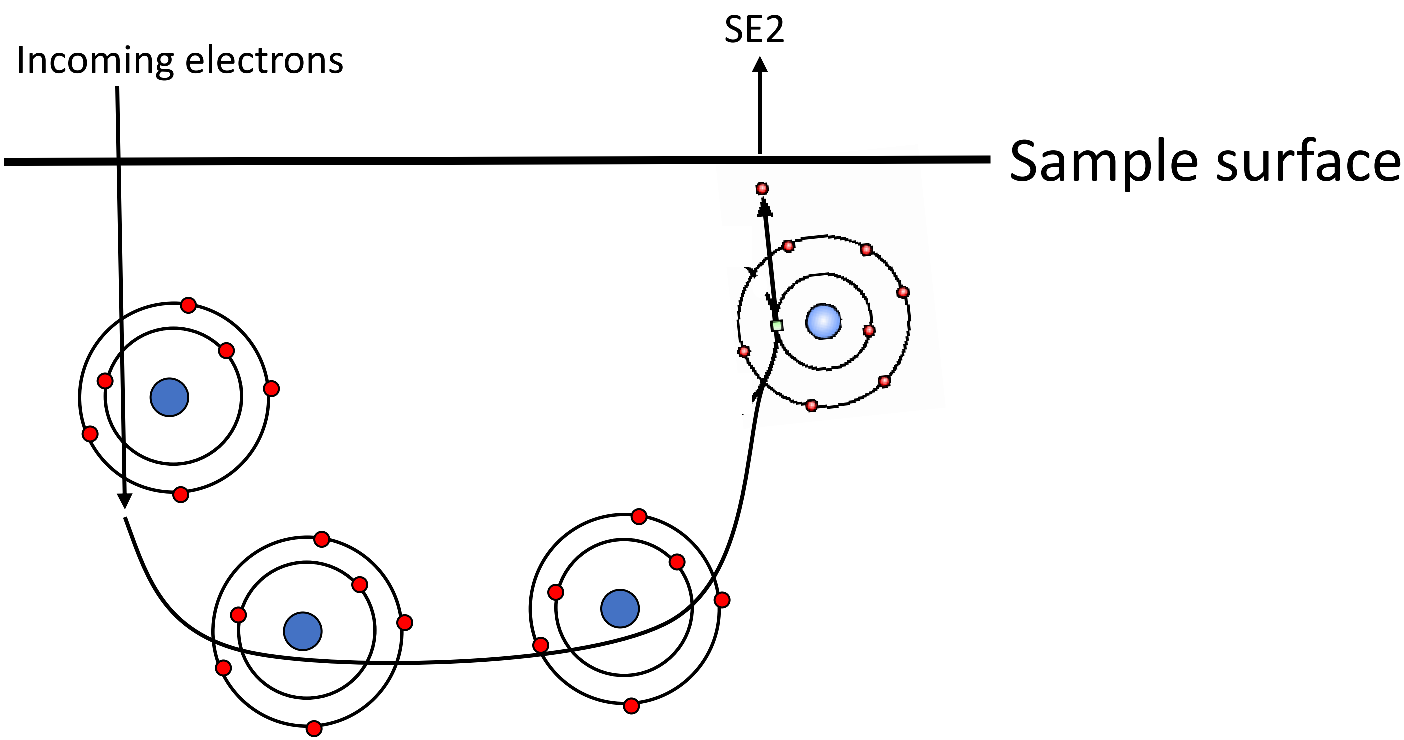
SE1

- The secondary electrons that are generated by the incoming electron beam as they enter the surface.
- High resolution signal with a resolution which is only limited by the electron beam diameter



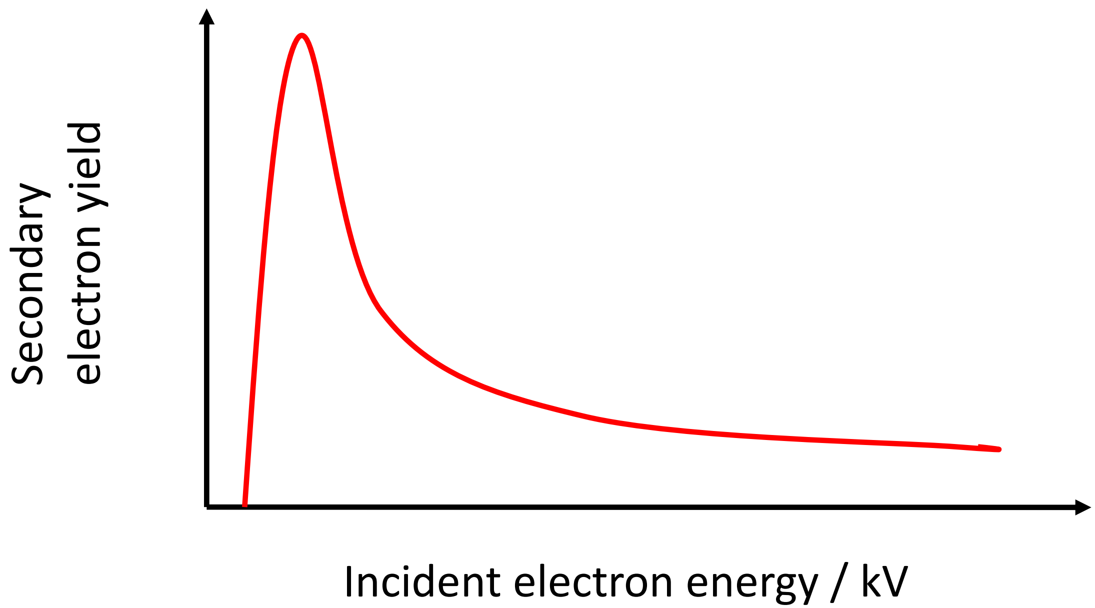
SE2

- Generated by the backscattered electrons that have returned to the surface after several inelastic scattering events.
- Resolution is poorer than SE1.



Factors that affect SE emission

1. Work function of the surface
2. Beam energy and beam current
   - Electron yield goes through a maximum at low acc. voltage, then decreases with increasing acc. voltage



Factors that affect SE emission

3. Atomic number (Z)
   - More SE2 are created with increasing Z
   - The Z-dependence is more pronounced at lower beam energies

4. The local curvature of the surface (the most important factor)

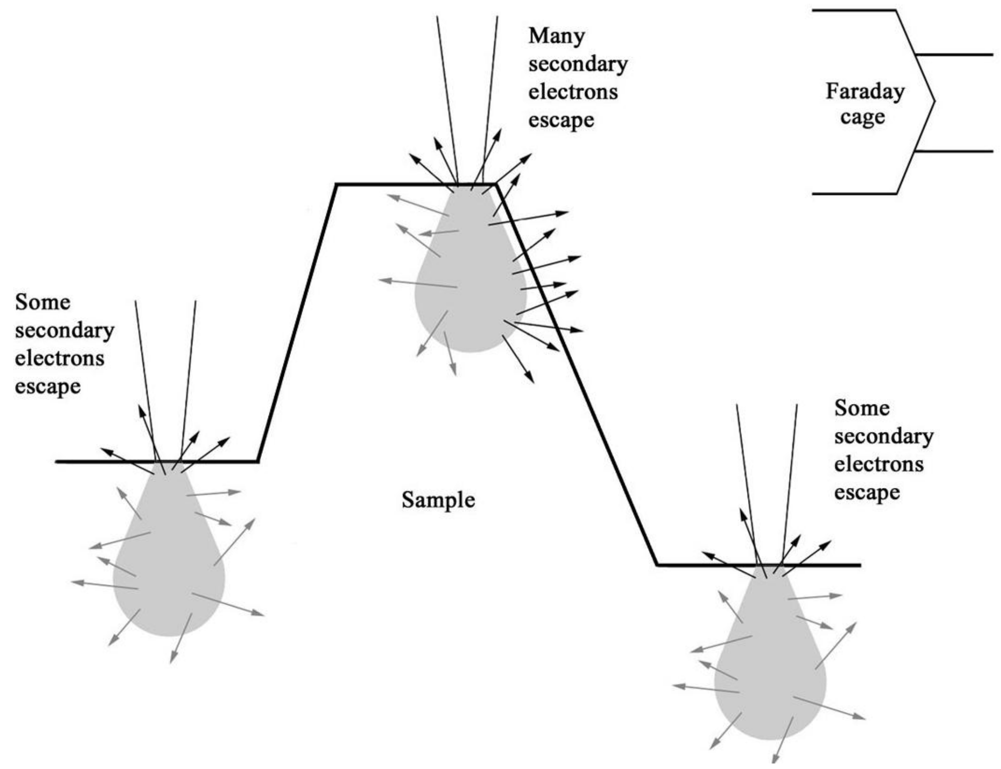
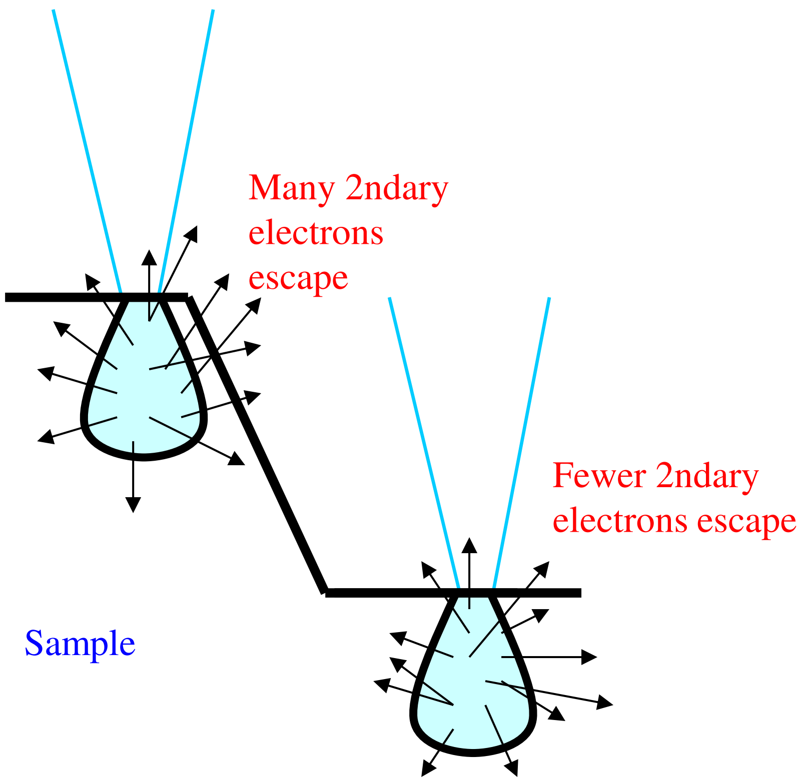


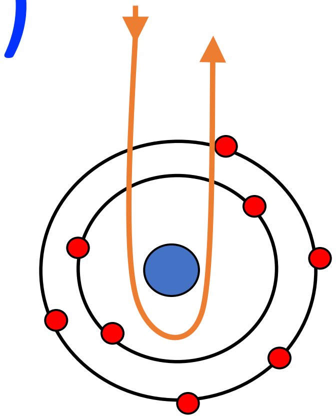
Image: Smith College Northampton, Massachusetts

Origin of topographic contrast in secondary electron images

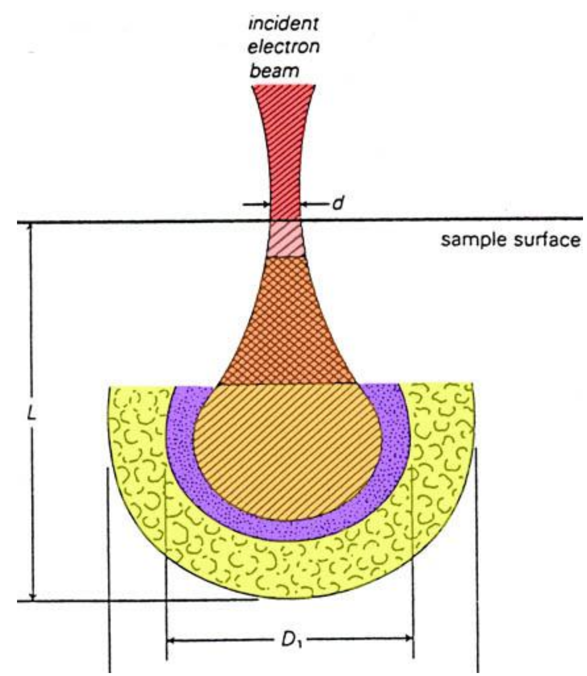


- The yield of secondary electrons is at a minimum when the surface of the specimen is perpendicular to the electron beam.
- At regions of the specimen which are not exactly perpendicular to the beam, electrons are more likely to be scattered out of the specimen.
- Hence, such regions appear bright in the secondary electron image.

Backscattered electrons (BSE)



- A fraction of the incident electrons is retarded by the electro-magnetic field of the nucleus and if the scattering angle is greater than 180 ° the electron can escape from the surface

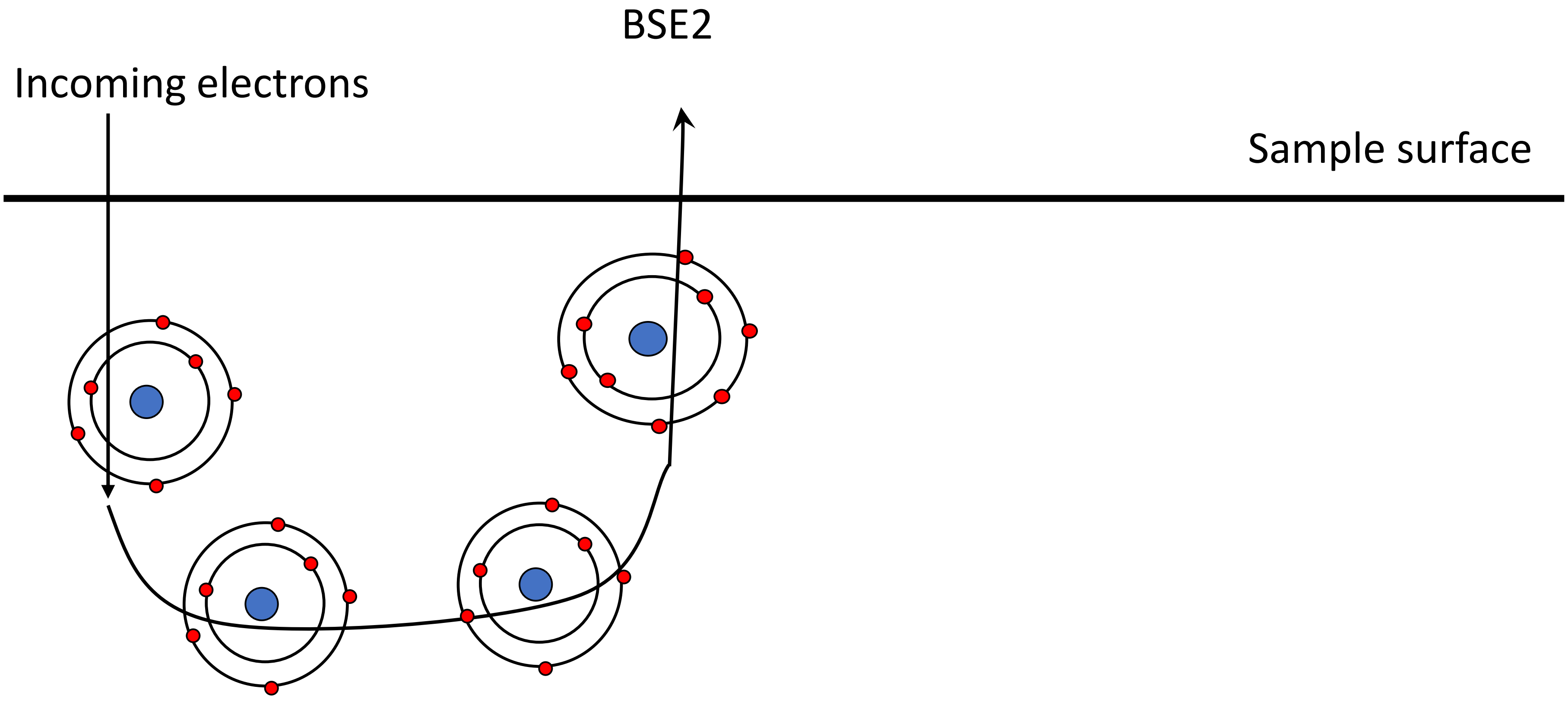


Backscattered electrons (BSE)

- High energy electrons (elastic scattering)
- Fewer BSE than SE
- We differentiate between BSE1 and BSE2
- Singly scattered backscattered electrons (BSE1) tend to emit at a high angle and are closely linked to compositional contrast,
- Multiply scattered BSE (BSE2) take off at a lower angle and are used to characterize composition and crystalline structures of a sample.

BSE2

- Most BSE are of BSE2 type



BSE as a function of atomic number

- Untuk fase yang mengandung lebih dari satu elemen, nomor atom rata-rata yang menentukan backscatter coefficient η

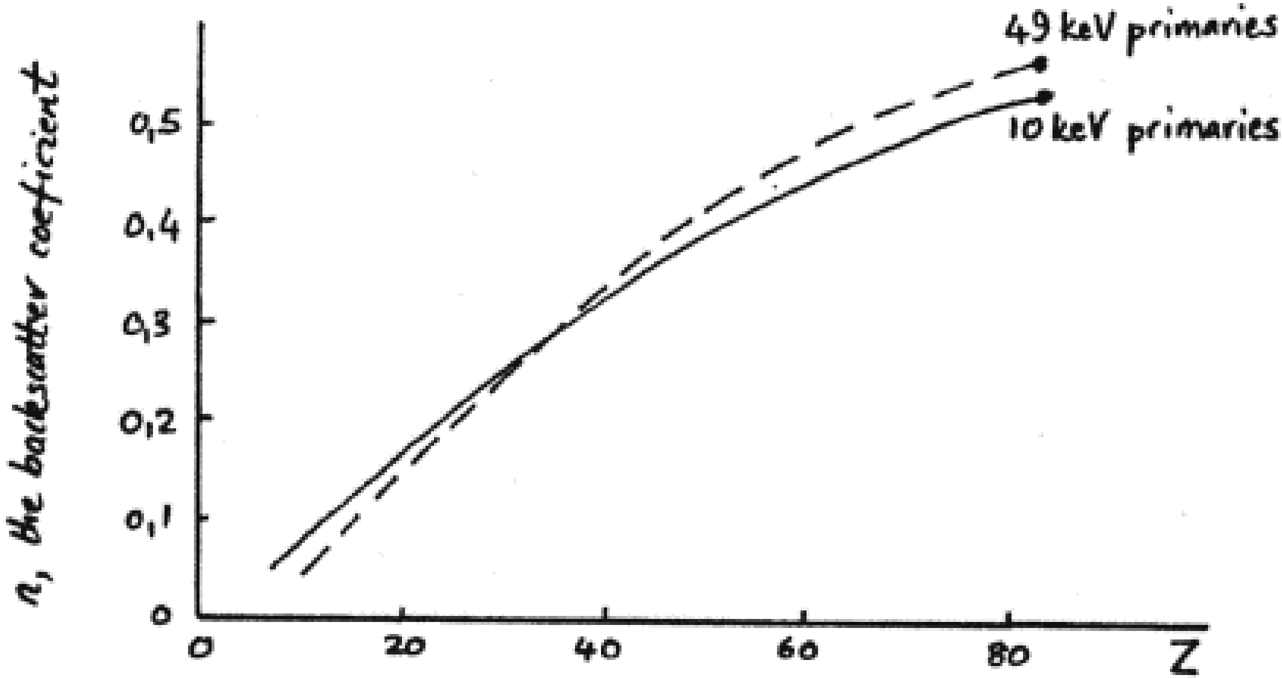


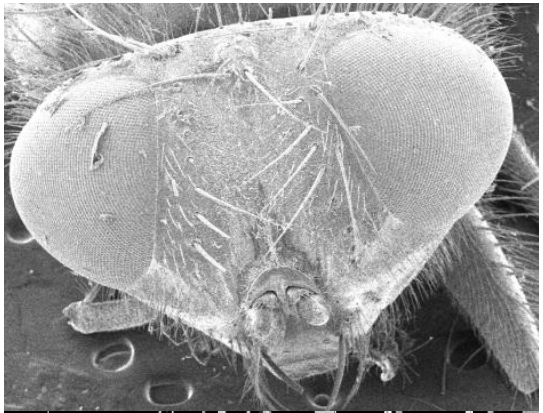
Image: University of Cape Town

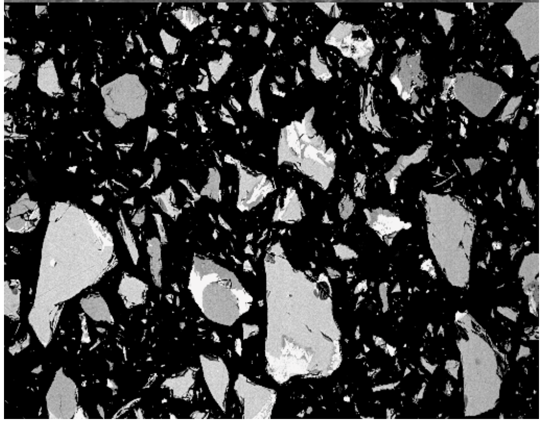
Factors that affect BSE emission

- Arah permukaan yang teriritasi
  - lebih banyak elektron akan mengenai detektor BSE ketika permukaan sejajar dengan detektor BSE
- Average atomic number
- Jika Anda ingin mempelajari perbedaan nomor atom, sampel harus dibuat setingkat mungkin (persiapan sampel!)

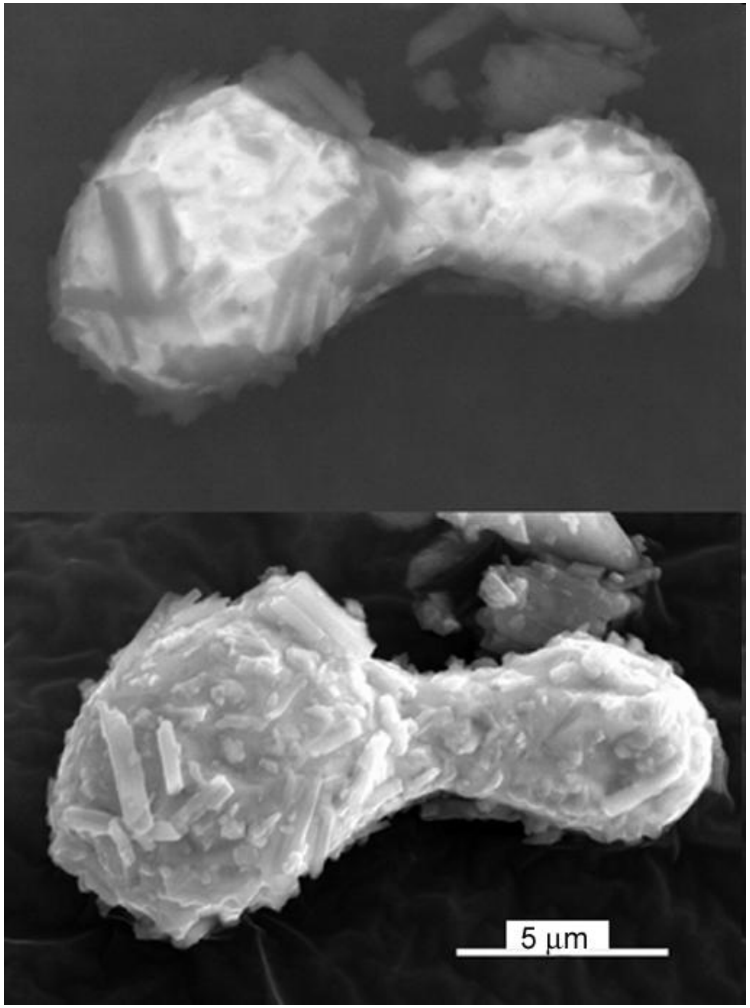
Information given by secondary and backscattered electrons

- Elektron sekunder berasal dari kedalaman beberapa nm dari permukaan.
- SE sangat sensitif terhadap struktur permukaan dan memberikan informasi topografi.
- BSE berasal dari jauh lebih dalam di dalam sampel (beberapa nm di bawah permukaan), dan berinteraksi lebih kuat dengan sampel.
- Oleh karena itu, **mereka memberikan informasi komposisi tetapi memberikan gambar dengan resolusi lebih rendah.**





BSE vs SE

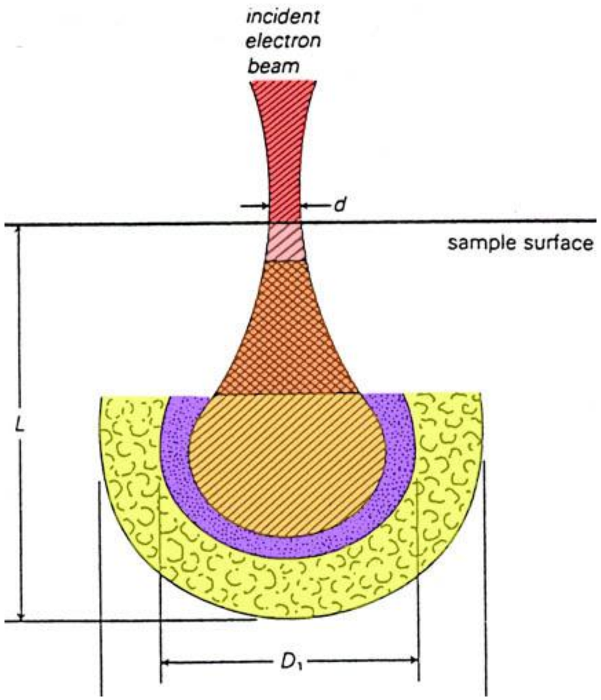


Images: Greg Meeker, USGS

X-rays

- *Photons* not electrons
- Each element has a *fingerprint* X-ray signal
- Poorer spatial resolution than BSE and SE
- Relatively few X-ray signals are emitted and the detector is inefficient

  → relatively long signal collecting times are needed



X-rays

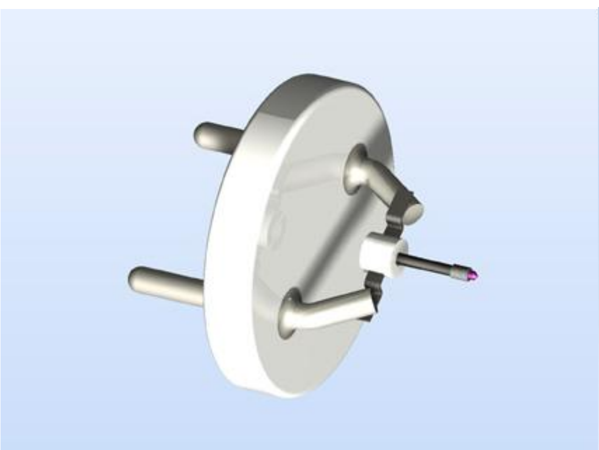
- Most common spectrometer: EDS (energy-dispersive spectrometer)
- Signal overlap *can* be a problem
- We can analyze our sample in different modes
  - spot analysis
  - line scan
  - chemical concentration map (elemental mapping)

Electron guns

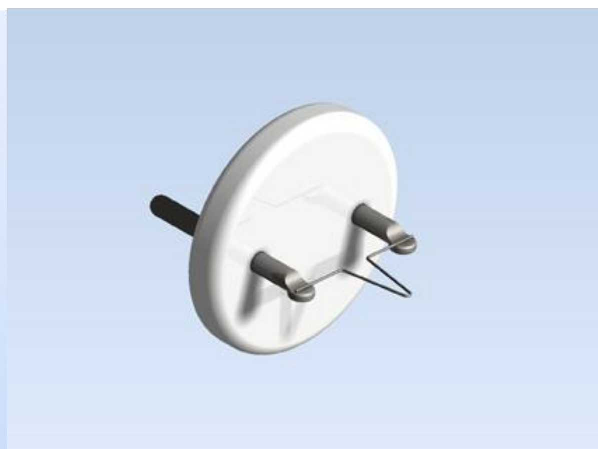
- We want many electrons per time unit per area (high current density) and as small electron spot as possible
- Traditional guns: thermionic electron gun (electrons are emitted when a solid is heated)
  - W-wire, LaB_6-crystal
- Modern: field emission guns (FEG) (cold guns, a strong electric field is used to extract electrons)
  - Single crystal of W, etched to a thin tip

Electron guns

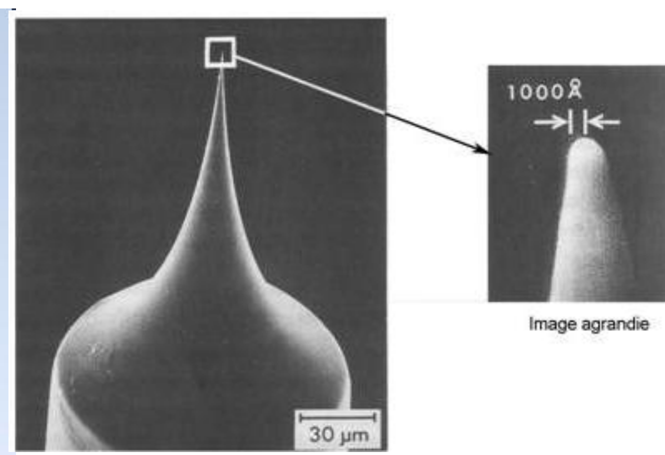
- With field emission guns we get a smaller spot and higher current densities compared to thermionic guns
- Vacuum requirements are tougher for a field emission guns



Single crystal of LaB_6



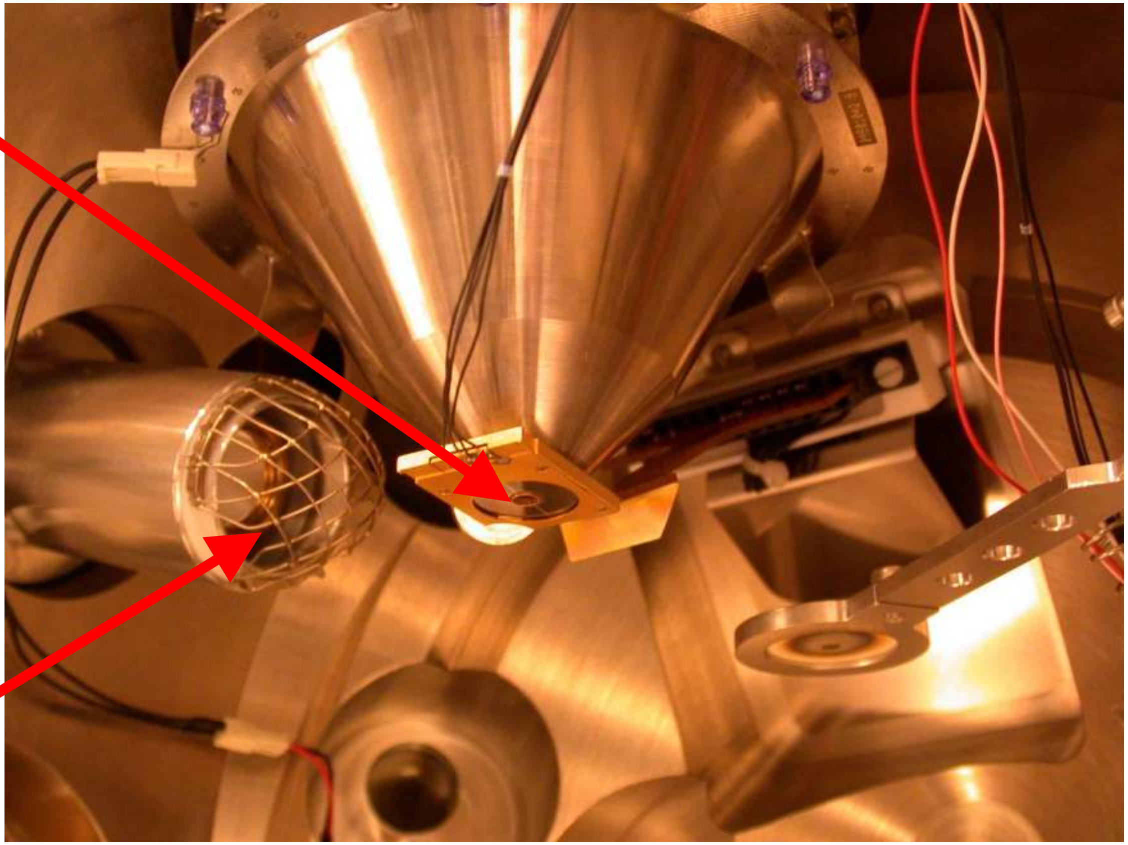
Tungsten wire



Field emission tip

Detectors

Backscattered electron detector:
(Solid-State Detector)



Secondary electron detector:
(Everhart-Thornley)

Image: Anders W. B. Skilbred, UiO

Our traditional detectors

- Secondary electrons: Everhart-Thornley Detector
- Backscattered electrons: Solid State Detector
- X-rays: Energy dispersive spectrometer (EDS)

Why do we need vacuum?

- Chemical (corrosion!!) and thermal stability is necessary for a well-functioning filament (gun pressure)
    - A field emission gun requires ~ 10^{-10} Torr
    - LaB_6: ~ 10^{-6} Torr
- The signal electrons must travel from the sample to the detector (chamber pressure)
    - Vacuum requirements is dependant of the type of detector

Environmental SEM: ESEM

- Traditional SEM chamber pressure: ~ 10^{-6} Torr
- ESEM: 0.08 – 30 Torr
- Various gases can be used
- Requires different SE detector

Why ESEM?

- To image challenging samples such as:
  - insulating samples
  - vacuum-sensitive samples (e.g. biological samples)
  - irradiation-sensitive samples (e.g. thin organic films)
  - “wet” samples (oily, dirty, greasy)
- To study and image chemical and physical processes in-situ such as:
  - mechanical stress-testing
  - oxidation of metals
  - hydration/dehydration (e.g. watching paint dry)

Some comments on resolution

- Best resolution that can be obtained: size of the electron spot on the sample surface
  - The introduction of FEG has dramatically improved the resolution of SEM's
- The volume from which the signal electrons are formed defines the resolution
  - SE image has higher resolution than a BSE image
- Scanning speed:
  - a weak signal requires slow speed to improve signal-to-noise ratio
  - when doing a slow scan drift in the electron beam can affect the accuracy of the analysis

Summary

- The scanning electron microscope is a versatile instrument that can be used for many purposes and can be equipped with various accessories
- An electron probe is scanned across the surface of the sample and detectors interpret the signal as a function of time
- A resolution of 1 – 2 nm can be obtained when operated in a high resolution setup
- The introduction of ESEM and the field emission gun have simplified the imaging of challenging samples

Summary

- Signals:
  - Secondary electrons (SE): mainly topography
    - Low energy electrons, high resolution
    - Surface signal dependent on curvature
  - Backscattered electrons (BSE): mainly chemistry
    - High energy electrons
    - “Bulk” signal dependent on atomic number
  - X-rays: chemistry
    - Longer recording times are needed

THANK YOU ☺